

Md Kamrujjaman Mobin

Software Engineer I, Samsung R&D Institute Bangladesh

kamrujjamanmobin123@gmail.com · +880-1749-500-360 · kmobin555.github.io · Google Scholar · LinkedIn · GitHub

Research Interests

Computer Vision, Embodied AI & Vision-Language Navigation, Vision-Language-Action (VLA) Models, Physical AI & Robotics, Multimodal Learning, Biomedical AI, Natural Language Processing.

Education

Shahjalal University of Science and Technology (SUST)

Sylhet, Bangladesh

B.Sc. in Computer Science and Engineering CGPA: 3.83 / 4.00

Feb 2020 – Jul 2025

Undergraduate Thesis: *AI-Driven Health System for Mortality Prediction Using Electrocardiograms (ECGs) with Transformers and Selective State-Space Models* [PDF]

Focused on deep sequence modeling, representation learning, and AI-driven healthcare analytics.

Professional Experience

Samsung R&D Institute Bangladesh

Dhaka, Bangladesh

Software Engineer I

July 2025 – Present

- **Brain Pulse App** (launching with Samsung Health, 2027): Android app for longitudinal cognitive health monitoring, correlating passive device-usage patterns with brain health biomarkers via on-device ML. Built multi-module Clean Architecture (Hilt DI, Room DB, WorkManager); passive pipelines for app usage, screen events, sleep, and call logs; on-device ONNX/TFLite inference over a 10+ biomarker feature extraction pipeline.
- **RSDS Mobile SDK** (Remote Sensor Data Service): Clinical-grade cross-device SDK enabling large-scale health research by collecting **40+ sensor modalities** from smartphone and Wear OS (PPG, ECG, EDA, BIA, accelerometer, sleep stages, app usage). KMP multi-module architecture with compile-time DI, SQLCipher-encrypted local storage, GZIP sync, and AES file encryption for HIPAA-compliant data handling.
- **Compass Watch App** (Galaxy Smartwatch 7 & 8): Maintained a system utility for wearable sensor calibration and validation. Improved codebase via unit testing and clean-code refactoring: **7.5% SAM score improvement, 20% CBO reduction, 90%+ unit test coverage**.
- Conducted applied R&D on **non-invasive blood glucose estimation** from PPG signals and multimodal **Alzheimer's risk modeling** — both targeting Samsung Health's **450M+ active user** platform.

Publications

2025

- [1] **Mobin, M. K.**, Islam, M. S., Al Barid, S., & Masum, M.
Cardioformer: Advancing AI in ECG Analysis with Multi-Granularity Patching and ResNet.
38th Canadian Conference on Artificial Intelligence (**Canadian AI 2025**), Calgary, Canada.
[arXiv:2505.05538](#)
- [2] **Mobin, M. K.** & Islam, M. S.
LuxVeri at GenAI Detection Task 1: Inverse Perplexity-Weighted Ensemble for Robust Detection of AI-Generated Text across English and Multilingual Contexts.
31st International Conference on Computational Linguistics (**COLING 2025**), Dubai, UAE.
[arXiv:2501.11914](#)
- [3] **Mobin, M. K.** & Islam, M. S.
LuxVeri at GenAI Detection Task 3: Cross-Domain Detection of AI-Generated Text Using Inverse Perplexity-Weighted Ensembles.
31st International Conference on Computational Linguistics (**COLING 2025**), Dubai, UAE.
[arXiv:2501.11918](#)

2024

- [4] **Mobin, M. K.**
Transforming ECG Images to Waveforms with UNet-LSTM and Multilabel Classification Using ResNet-LSTM: A Multimodal Approach.
51st Computing in Cardiology (**CinC 2024**), Karlsruhe, Germany.
cinc.org/archives/2024/pdf/CinC2024-393.pdf

- [5] Mobin, M. K., Al Barid, S., & Rahman, M. S.
CricPose360: A Comprehensive Dataset for Real-Time Pose-Enhanced Cricket Shot Classification Using YOLOv8.
 27th International Conference on Computer and Information Technology (ICIT 2024), Bangladesh.
 DOI: 10.1109/ICIT64611.2024.11022536

Research Projects

- **Event-Based Vision & ANN–SNN Hybrid Perception** · *Neuromorphic Vision, Robotics*. Jan 2026 – Now
 Surveyed 10+ papers (EventFly CVPR 2025, HsVT, SSM-based event models) and ran hands-on experiments on N-EPIC Kitchens: voxel-grid generation, E2VID reconstruction [NB1][NB2], and SSM+SNN hybrid temporal modeling [NB3]. Proposing spike-native state-space models and noise-aware ANN–SNN hybrids for cross-platform event perception.
- **GSA-VLN: General Scene Adaptation for VLN** · *Embodied AI, ICLR 2025 Replication* · [GitHub](#) 2026
 Replicated GR-DUET (ICLR 2025) on R2R/Matterport3D. Implemented two novel improvements: **Semantic-Aware Navigation** with InfoNCE contrastive loss grounding instructions to room-type-matching graph nodes (+1.4% SR, +1.6% SPL), and **Experience Replay** for continual learning across scenes (+0.7% late-episode SR).
- **MindCube Spatial Reasoning Challenge** · *VLM Fine-tuning & Ensembling* · [Challenge](#) · [EvalAI](#) 2026
 Fine-tuned Qwen2.5-VL-3B-Instruct via QLoRA on MindCube using structured cognitive-map + chain-of-thought outputs to force explicit spatial reasoning before prediction. Built a 3-run weighted ensemble of a custom SFT model and the official GRPO-trained RL model with self-consistency sampling for final answer aggregation.
- **Kaggle Competitions** 2022 – Present
 - **Image Matching Challenge 2024** — **Bronze Medal** (Top 6.3%): Built a multi-model 3D reconstruction pipeline integrating COLMAP with deep feature matchers. [Notebook](#)
 - **ISIC 2024: Skin Cancer Detection** — **Bronze Medal** (Top 5.7%): Developed CNN-based multimodal models with advanced augmentations for melanoma classification. [Notebook](#)

Ongoing Research

- **Event-Based Multimodal Understanding Challenge** — **Active Participant**: General Understanding Track at ECCV 2026 Workshop on Event-Based & Multimodal Vision (EBMV); developing event-stream MLLM solutions for spatiotemporal reasoning. [Competition](#)
- **Serialization Bottleneck in Tool-Use Agents**: Measuring how text-serialized intermediate representations (polygons, graphs, tables, ASTs) degrade compositional LLM agent pipelines; evaluating runtime-managed object handles with deterministic property summaries as mitigation.

Technical Skills

Programming	Python, C++, Java, Kotlin, JavaScript, C
ML / AI Frameworks	PyTorch, Hugging Face Transformers, Scikit-learn, OpenCV, NumPy, Pandas
Research Areas	Computer Vision, NLP, Multimodal Learning, Embodied AI, VLA, Signal Processing
Academic	Research Writing, Experimental Design, Scientific Presentation, Peer Review

Language Proficiency

IELTS Academic: 7.0

November, 2025

Listening: 7.5 | Reading: 7.0 | Speaking: 6.5 | Writing: 6.0

References

Md Saiful Islam

Data Scientist

Dept. of Computer Science

Athabasca University, Canada

saiful-cse@sust.edu | [Google Scholar](#) | [LinkedIn](#)

Prof. Mohammad Shahidur Rahman, PhD

Professor, Computer Science & Engineering

Shahjalal University of Science & Technology,

Bangladesh

rahmanms@sust.edu | [Google Scholar](#) | [LinkedIn](#)

Prof. Md Masum

Professor & Department Head, Computer Science & Engineering

Shahjalal University of Science & Technology, Bangladesh

masum-cse@sust.edu | [Personal Page](#) | [IEEE Xplore](#)